

Rodolfo Ostilla Mónico

Associate Professor (Profesor Titular), Fluid Mechanics

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Summary

Fluid dynamicist specializing in the direct numerical simulation of turbulent flows, with research spanning thermal convection and sustainable ventilation, the hydrodynamics of musical instruments, shear and vortex flows, and the physics of watersports. According to Clarivate/WoS, an h-index of 22 and over 1,400 citations across 59 peer-reviewed articles, including publications in Physical Review Letters, Journal of Fluid Mechanics, Science Advances, and PNAS. Previously a postdoctoral researcher at Harvard University and Assistant Professor at the University of Houston; now an Emergia fellow and Associate Professor at the Universidad de Cádiz.

Education

PhD	Twente University , Physics of Fluids	Enschede, Overijssel, Netherlands
	- Thesis: Highly turbulent Taylor-Couette flow: direct numerical simulations - Advisors: D. Lohse (director), R. Verzicco (co-director)	Jan 2011 – Feb 2015
MSc	Cranfield University , Aerospace Dynamics	Sept 2010
Ingeniero Aeronáutico (5-year degree)	Universidad de Sevilla , Aeronautical Engineering	Sept 2010
Grado	Universidad Nacional de Educación a Distancia (UNED) , Philosophy	June 2022

Experience

Universidad de Cádiz , Assistant Professor / Profesor Titular	Escuela Superior de Ingeniería, Puerto Real, Spain
Department of Mechanical Engineering and Industrial Design, School of Engineering	July 2026 – present 1 month
Universidad de Cádiz , Emergia Postdoctoral Fellow	Escuela Superior de Ingeniería, Puerto Real, Spain
	Mar 2022 – July 2026 4 years 5 months
University of Houston , Assistant Professor	Cullen College of Engineering, Houston, TX, USA
- Taught undergraduate and graduate courses in fluid mechanics and non-linear dynamics - Supervised 5 MSc theses, 4 undergraduate research projects, and 1 PhD thesis	Oct 2017 – Feb 2021 3 years 5 months
Harvard University , Postdoctoral Researcher	Cambridge, MA, USA
	Oct 2015 – Sept 2017 2 years

European Space Astronomy Centre (ESAC), Trainee

Madrid, Spain
Oct 2010 – Mar 2011
6 months

Universidad de Sevilla, Research Assistant

Departamento de Mecánica
de Fluidos, Seville, Spain
Jan 2008 – Jan 2009
1 year 1 month

Projects

Aeroacústica e hidrodinámica de los instrumentos musicales tipo chimenea

Sept 2023 – Aug 2027

University of Cádiz, competitive research project funded by the Spanish State Research Agency (Proyectos de Generación de Conocimiento 2022). Main PI.

- Total funding: €81,250

CDS&E: Multi-scale Coherent Structure Extraction and Tracking for Modern CFD Data Analysis

Sept 2021 – Aug 2024

University of Houston, competitive research project funded by the U.S. National Science Foundation. Co-PI with Guoning Chen and Di Yang.

- Total funding: \$453,000

Particulate Rayleigh-Bénard Flow

Sept 2021 – Aug 2024

University of Houston, competitive research project funded by the U.S. National Science Foundation. Principal Investigator.

- Total funding: \$209,666

Controlling Secondary Flows by the Use of Non-Wetting Surfaces

Sept 2019 – Aug 2022

University of Houston, competitive research project funded by the U.S. National Science Foundation. Co-PI with Kamran Alba.

- Total funding: \$340,000

Awards

Emergia Fellowship — Agencia Andaluza del Conocimiento / Junta de Andalucía

Jan 2022

Fellowship funding return to Spain and research at the Universidad de Cádiz (€85,000).

Publications

Elastic turbulence in space

Jan 2025

Rodolfo Ostilla-Mónico

doi.org/10.1017/jfm.2024.1214 (Journal of Fluid Mechanics)

Fluid dynamical pathways of airborne transmission while waiting in a line

Jan 2025

Ruixi Lou, Milo Van Mooy, Gabriel A. Tarditti, Rodolfo Ostilla-Mónico, Varghese Mathai

doi.org/10.1126/sciadv.adw0985 (Science Advances)

Transition between boundary-limited scaling and mixing-length scaling of turbulent transport in internally heated convection

Jan 2022

Sina Kazemi, Rodolfo Ostilla-Mónico, David Goluskin

doi.org/10.1103/PhysRevLett.129.024501 (Physical Review Letters)

Controlling secondary flows in Taylor-Couette flow using axially spaced superhydrophobic surfaces

Jan 2023

Vignesh Jegannathan, Tala Shannak, Kamran Alba, Rodolfo Ostilla-Mónico

doi.org/10.1017/jfm.2023.606 (Journal of Fluid Mechanics)

Instability and disintegration of vortex rings during head-on collisions and wall interactions Aakash Mishra, Alain Pumir, Rodolfo Ostilla-Mónico doi.org/10.1103/PhysRevFluids.6.104702 (Physical Review Fluids)	Jan 2021
Turbulence generation through an iterative cascade of the elliptical instability Ryan McKeown, Rodolfo Ostilla-Mónico, Alain Pumir, Michael P. Brenner, Shmuel M. Rubinstein doi.org/10.1126/sciadv.aaz2717 (Science Advances)	Jan 2020
Physical and geometric constraints shape the labyrinth-like nasal cavity David Zwicker, Rodolfo Ostilla-Mónico, Daniel E. Lieberman, Michael P. Brenner doi.org/10.1073/pnas.1714795115 (Proceedings of the National Academy of Sciences)	Jan 2018
Logarithmic mean temperature profiles and their connection to plume emissions in turbulent Rayleigh-Bénard convection Erwin P. van der Poel, Rodolfo Ostilla-Mónico, Roberto Verzicco, Siegfried Grossmann, Detlef Lohse doi.org/10.1103/PhysRevLett.115.154501 (Physical Review Letters)	Jan 2015
A pencil distributed finite difference code for strongly turbulent wall-bounded flows Erwin P. van der Poel, Rodolfo Ostilla-Mónico, John Donners, Roberto Verzicco doi.org/10.1016/j.compfluid.2015.04.007 (Computers & Fluids)	Jan 2015
Optimal Taylor-Couette flow: direct numerical simulations Rodolfo Ostilla, Richard J. A. M. Stevens, Siegfried Grossmann, Roberto Verzicco, Detlef Lohse doi.org/10.1017/jfm.2012.596 (Journal of Fluid Mechanics)	Jan 2013

Teaching

Universidad de Cádiz , Profesor Titular	Escuela Superior de Ingeniería
Courses: Mecánica de Fluidos II (Grado en Ingeniería Aeroespacial). Supervised 11 undergraduate final-year projects (co-supervised 1); currently supervising 2 PhD students (co-supervising 1).	Jan 2022 – Jan 2025
University of Houston , Assistant Professor	Cullen College of Engineering
Mechanical Engineering BSc and MSc courses	Jan 2017 – Jan 2022

Skills

Computational Fluid Dynamics: Direct Numerical Simulation, Large-Eddy Simulation, GPU-accelerated solvers, turbulence, Taylor-Couette flow, Rayleigh-Bénard convection
Applied Fluid Mechanics: Aeroacoustics, vortex dynamics, ventilation and airborne transmission, hydrodynamics of musical instruments, biomechanics of flow

Languages

Spanish
Native speaker

English
C2

Mandarin Chinese
B1